TEST 9 INNOVATOR
MATH MODULE 1

1.

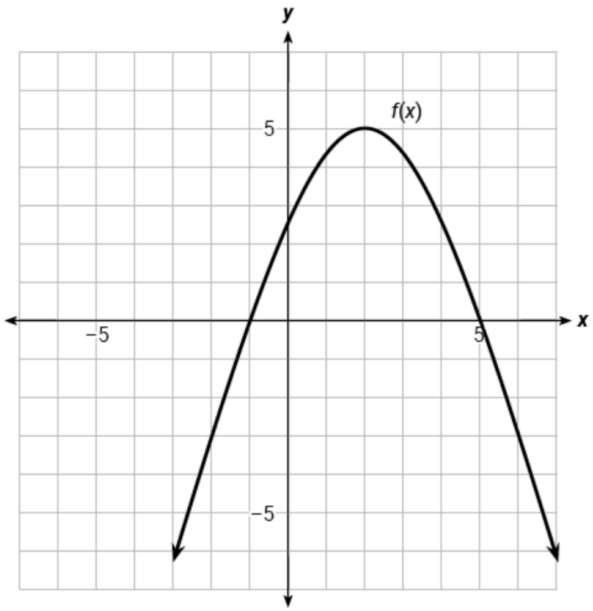
$$3(x-2)+5y=2$$
$$y=-2(x-5)$$

The solution to the given system of equations is (x, y). What is the value of y?

NEXT QUESTION

COME BACK LATER

2.



The graph of the function $f(x)$ is shown. What value(s) of x satisfy $f(x)=0$.

PREV. QUESTION

NEXT QUESTION

A 2.5

B 3.5

C -1 and 5

D There are no values of x where $f(x)=0$.

COME BACK LATER

3. Which of the following points satisfies the equation $4x - 9y = -1$?

PREV. QUESTION | NEXT QUESTION

- A $(-2, 1)$
- B $(7, 3)$
- C $(9, 4)$
- D $(11, 5)$

4. The function $g(x) = kx + 2$, where k is a constant, is graphed on the xy-plane and passes through the coordinate point $(5, -8)$. What is the value of k?

PREV. QUESTION | NEXT QUESTION

- A -2
- B $-\frac{6}{5}$
- C $-\frac{3}{8}$
- D 2

5. $5x^2 + 16 = 41$

Which of the following is a solution to the given equation?

PREV. QUESTION | NEXT QUESTION

× A $\sqrt{5}$

× B $\sqrt{\frac{57}{5}}$

× C 5

× D 625

6. Which expression is equivalent to $(3x^2 - 3x) + (4x^2 - 8x)$?

PREV. QUESTION | NEXT QUESTION

× A $-4x^2$

× B $7x^2 - 11x$

× C $7x^4 - 11x^2$

× D $12x^4 + 24x^2$

7. If the expression $8x^2 - 18$ is equivalent to $a(4x^2 - 9)$, what is the value of a?

PREV. QUESTION | NEXT QUESTION

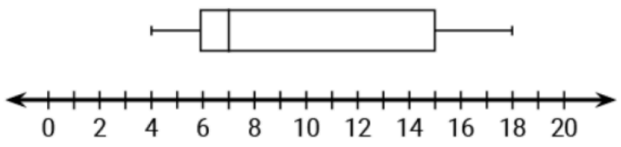
× A 2

× B 4

× C 8

× D 9

8.



The number of minutes it took for people to complete a survey is represented in the box plot shown. What is the median amount of time, in minutes, it took for people to complete the survey?

PREV. QUESTION | NEXT QUESTION

× A 6

× B 7

× C 9

× D 15

9. The expression $(ax + 2)(bx - 3)$ is equivalent to the expression $4x^2 + 5x - 6$. What is the product of a and b?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

10. $$(x + 4)(x - 3)(x - 9) = 0$$

What is the sum of the solutions to the given equation?

PREV. QUESTION | NEXT QUESTION

× A -16

× B -8

× C 8

× D 16

11. Let $f(x) = 3x - 7$ and $g(x) = -2x + 5$. What is the value of $3f(4) - g(4)$?

PREV. QUESTION | NEXT QUESTION

A -3

B 12

C 15

D 18

12. $$f(x) = x^2 - 9x - 22$$

If the given function $f(x)$ is graphed in the xy-plane, where $y = f(x)$, what is the x-coordinate of an x-intercept of the graph?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

13.



The scatterplot shows the comparison for the time and the cost to hire a personal trainer. Which of the following is the best approximate for the slope of the line of best fit?

PREV. QUESTION | NEXT QUESTION

- A 25
- B 75
- C 150
- D 200

COME BACK LATER

14. A small bakery shop sells only brownies and cupcakes. Each cupcake costs $\$5$ and each brownie costs $\$3$. On a particular day, the shop sold a total of 140 baked goods and earned a total of $\$536$. If b is the number of brownies sold and c is the number of cupcakes sold, which of the following systems of equations models this scenario?

PREV. QUESTION | NEXT QUESTION

- A $3b + 5c = 140$, $b + c = 536$
- B $3b + 5c = 536$, $b + c = 140$
- C $5b + 3c = 140$, $b + c = 536$
- D $5b + 3c = 536$, $b + c = 140$

15.

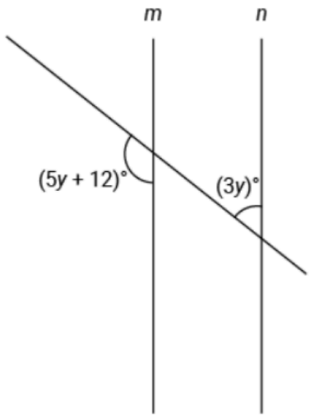
Hour	Ian	Kari
1	4	3
2	9	8
3	14	13
4	19	18

Ian and Kari are working together sewing patches on a quilt. The table shows how many patches they sewed on a quilt after a certain amount of hours. Which of the following statements is true?

PREV. QUESTION | NEXT QUESTION

- A. There will be a time when Kari and Ian will have sewn the same number of patches.
- B. Ian sews patches at a faster rate than Kari.
- C. A quilt with 77 patches will be completed in 7 hours.
- D. Kari and Ian sew patches at the same rate.

16.



Note: Figure not drawn to scale

In the figure shown, line m is parallel to line n. What is the value of y?

PREV. QUESTION | NEXT QUESTION

- A. 6
- B. 14
- C. 21
- D. 24

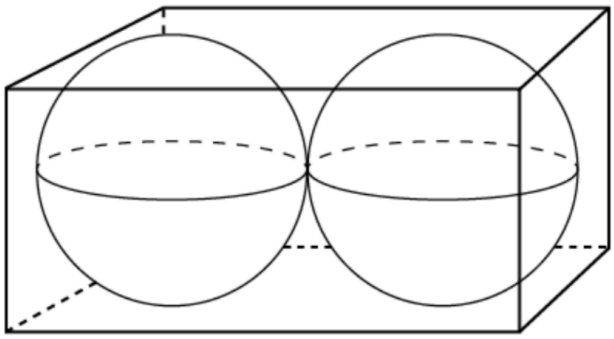
COME BACK LATER

17. A circle in the xy-plane has its center at $(-2, 5)$. Line l is tangent to the circle at the point $(1, 11)$. What is the slope of line l?

PREV. QUESTION | NEXT QUESTION

- A -2
- B $-\frac{1}{2}$
- C $\frac{1}{2}$
- D 2

18.



Two identical spheres, each with a volume of 972π cubic inches, fit perfectly inside a rectangular prism, as shown. What is the volume of the rectangular prism, in cubic inches?

PREV. QUESTION | NEXT QUESTION

- A $1{,}458$
- B $2{,}916$
- C $5{,}832$
- D $11{,}664$

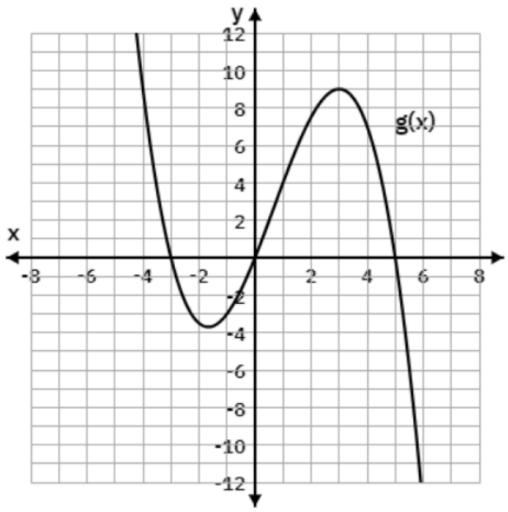
COME BACK LATER

19. In a random sample of n people, it is determined that p percent of people own a car. If this sample was taken from a population that totals $4n$ people, the expected percentage of people who own a car is represented by kp, where k is a rational number. What is the value of k?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

20.



The graph of $y = g(x)$ is shown on the xy-plane. If a is a constant and $a > 0$, which of the following could be the equation of $g(x)$?

PREV. QUESTION | NEXT QUESTION

A $g(x) = -a(x)(x-3)(x+5)$

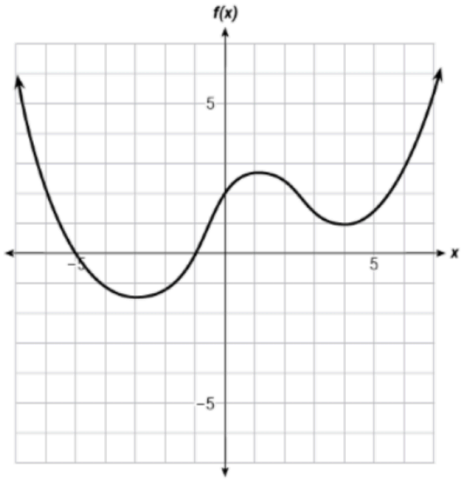
B $g(x) = -a(x)(x+3)(x-5)$

C $g(x) = a(x)(x-3)(x+5)$

D $g(x) = a(x^2)(x+3)(x-5)$

COME BACK LATER

21.



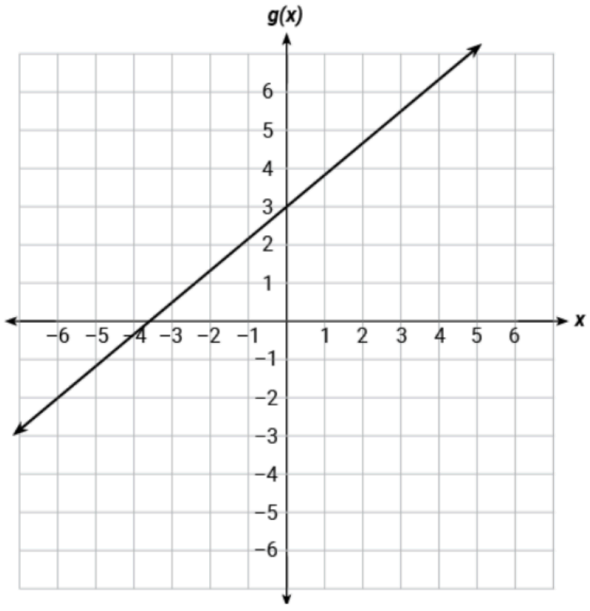
PREV. QUESTION | NEXT QUESTION

- A $(x-5)$
- B $(x-4)$
- C $(x+1)$
- D $(x+2)$

COME BACK LATER

The polynomial function $f(x) = ax^4 + bx^3 + cx^2 + dx + e$, where a, b, c, d, and e are constants, is graphed on the coordinate plane. If $f(x)$ is written in factored form, which of the following must be one of the factors of $f(x)$?

22.



PREV. QUESTION

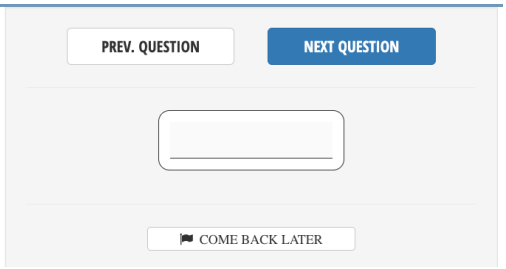
- A $g(x) = \frac{5}{6}x - 5$
- B $g(x) = \frac{5}{6}x - 2$
- C $g(x) = \frac{5}{6}x + 3$
- D $g(x) = \frac{5}{6}x + 8$

COME BACK LATER

The graph of the function $g(x) - 5$ is shown. Which equation defines the function of $g(x)$?

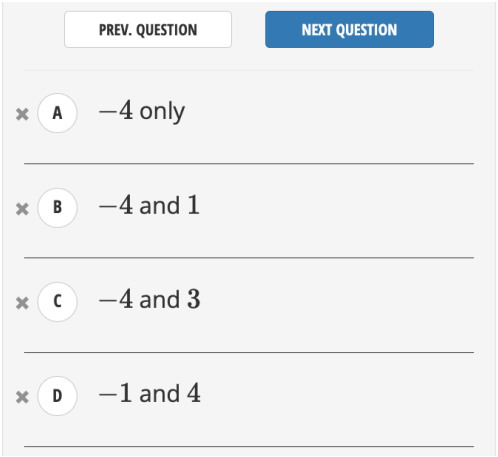
MODULE 2B (HARDER)

1. If 80% of a number is 104, what is 40% of the same number?

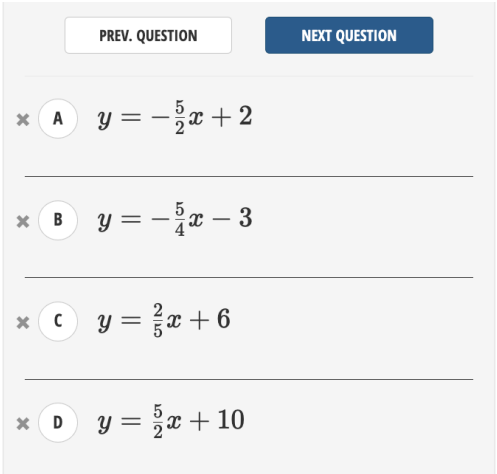


2. $$g(n) = n^2 + 3n - 4$$

The function $g(n)$ is defined for all values of n. What values of n satisfy $g(n) = 0$?



3. When graphed in the xy-plane, line m passes through the points $(-5, -4)$ and $(15, 4)$. If line n is perpendicular to line m, which of the following could be the equation of line n?



4.

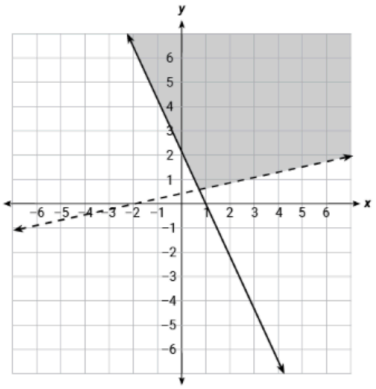
$$y > \frac{1}{5}x + \frac{1}{2}$$

$$y \leq -2x + 2$$

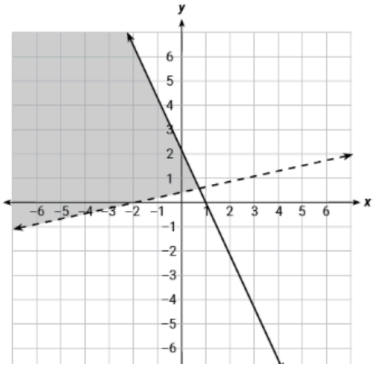
The given system of inequalities is graphed in the xy–plane. Which graph shows the solution set for the inequality?

PREV. QUESTION NEXT QUESTION

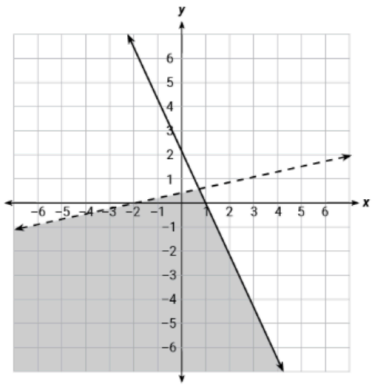
A



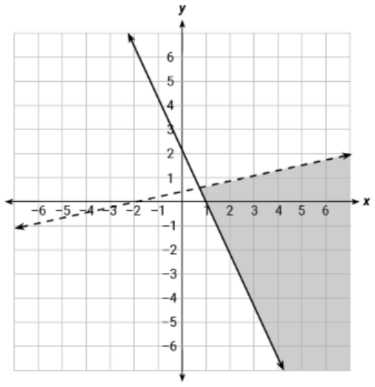
B



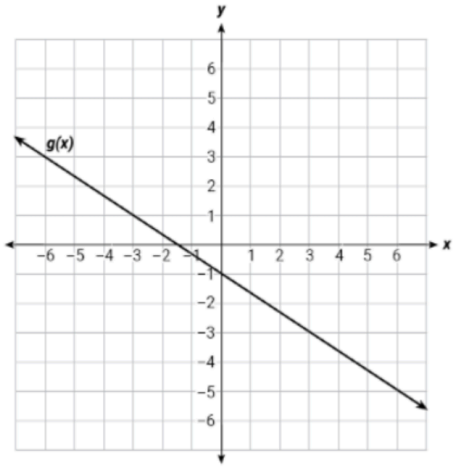
C



D



5.



The graph of the function $g(x)$ is shown.
If $h(x) = g(x) + 4$, what is the value of $h(6)$?

PREV. QUESTION

NEXT QUESTION

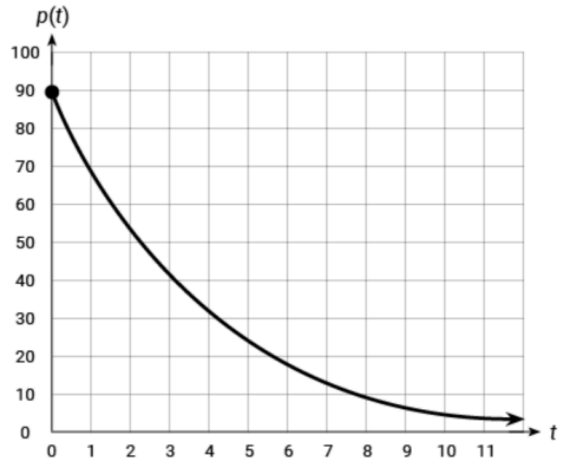
COME BACK LATER

6. The function $h(t) = -4.9t^2 + 14.7t + 19.6$ determines the height of a cannonball from the ground, in meters, t seconds after it is fired from a cannon over a cliff. How long, in seconds, is the cannonball in the air before it touches the ground?

PREV. QUESTION | NEXT QUESTION

- A 1
- B 4
- C 14.7
- D 19.6

7.



The function $p(t)$ gives the population of bees, in thousands, in a park over t years since data was first collected in 2008. The function is graphed on the coordinate plane. Which statement is the best interpretation of where the graph intersects the $p(t)$ axis?

PREV. QUESTION | NEXT QUESTION

- A There were 90 bees in the park in 2008.
- B The total number of bees between 2008 and 2019 decreased by 90.
- C There were $90{,}000$ bees in the park in 2008.
- D There were $90{,}000$ bees in the park in 1997.

COME BACK LATER

8. The circumference of a circle is equal to $w\pi$ feet, and the area of a circle is equal to $w\pi$ square feet, where w is a constant. What is the value of w?

PREV. QUESTION | NEXT QUESTION

- A 1
- B 2
- C 4
- D 8

9. $$4^2 = 2^{x-4}$$

What value of x satisfies the given equation?

PREV. QUESTION | NEXT QUESTION

- A 4
- B 6
- C 8
- D 10

10. $$3x^2 - 6x - 11 = 0$$

If one of the solutions to the given quadratic equation is $\frac{3+\sqrt{g}}{3}$, what is the value of g?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

11. $$27x^2 + kx + 12 = 0$$

In the given equation, k is a constant. For which of the following values of k will the equation have no real solutions?

PREV. QUESTION | NEXT QUESTION

A -55

B -36

C 34

D 40

12. A high school gave a survey to all of its students. It was determined that 28% of the students are ninth graders. Out of all the ninth graders, 15% walk to school. If 21 ninth graders walk to school, how many students attend the high school in total?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

13. $$9x - 8y = -26$$
$$5x - 6y = -30$$

The coordinate point (x, y) is the solution to the system of equations shown. What is the value of $x + y$?

PREV. QUESTION | NEXT QUESTION

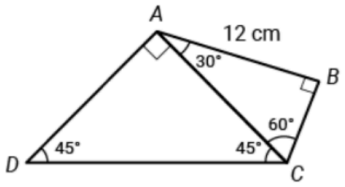
A 4

B 6

C 10

D 16

14.



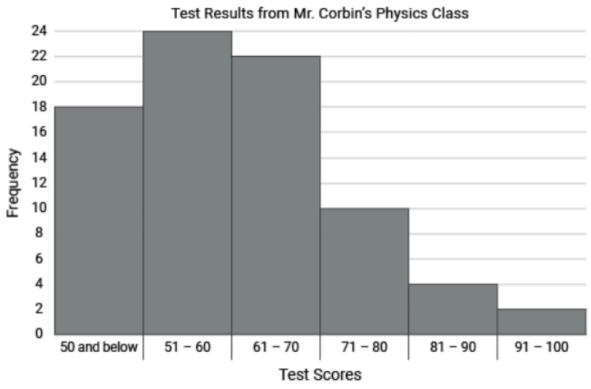
Note: Figure not drawn to scale.

In the figure shown, what is the length of side CD, in centimeters?

PREV. QUESTION | NEXT QUESTION

- A $4\sqrt{6}$
- B $8\sqrt{3}$
- C $6\sqrt{6}$
- D $8\sqrt{6}$

15.



Mr. Corbin recently had his students take a physics test. He made a mistake on one of the test questions, so Mr. Corbin added 10 points to every student's score. The histogram summarizes the test scores after adding the 10 extra points. In which range is the median score before the 10 points were added?

PREV. QUESTION | NEXT QUESTION

- A 50 and below
- B $51 - 60$
- C $61 - 70$
- D $71 - 80$

COME BACK LATER

16. In the xy-plane, the graph of $y = 4x^2 - 48x + 44$ intersects the line $y = b$ at exactly one point. What is the value of b?

PREV. QUESTION | NEXT QUESTION

A) -100

B) -1

C) 6

D) 11

17. $$\frac{2x^2}{x^2-4} + \frac{7x}{x+2} = \frac{-2}{x-2}$$

What value of x satisfies the given equation?

PREV. QUESTION | NEXT QUESTION

A) -2

B) $-\frac{2}{3}$

C) $\frac{2}{3}$

D) 2

18. A school budgets $\$1{,}600$ to buy pens. A bulk order of pens requires a minimum purchase of at least 520 pens for a discounted price. The store charge $\$2$ per black pen and $\$4$ per blue pen. What is the maximum number of blue pens the school can order to stay within budget and receive the discounted price?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

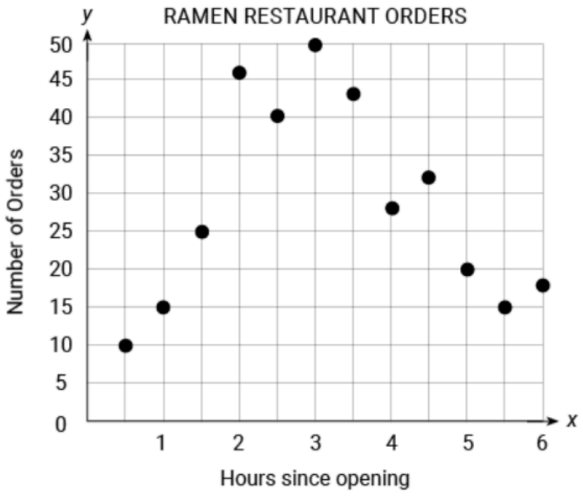
19. $0.84y^2 - 7.2y + 1.56$

The given expression is equivalent to $k(-7y^2 + 60y - 13)$, where k is a constant. What is the value of k?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

20.



A ramen restaurant tracked how many orders they received during the first 6 hours of opening. The scatterplot shows the results. Which of the following equations best models the data in the scatterplot?

PREV. QUESTION | NEXT QUESTION

A $y = -4.21x^2 + 26.97x - 2.09$

B $y = -4.21x^2 - 26.97x - 2.09$

C $y = 4.21x^2 + 26.97x + 2.09$

D $y = 4.21x^2 - 26.97x + 2.09$

COME BACK LATER

21. Circle A is centered at the origin with a radius of R. Circle B has a radius of r such that $r < R$, is centered at the point (r, r), and intersects circle A at exactly one point. What is the value of R in terms of r?

PREV. QUESTION | NEXT QUESTION

- A $2r$
- B $2r\sqrt{2}$
- C $r(\sqrt{2}+1)$
- D $r(\sqrt{2}+2)$

22. $$b^{\frac{4}{5}} = c^{\frac{2}{3}}$$

Given the equation above, if $c = b^{2a+1}$, then what is the value of a?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER